

Readme Contents

1. What is this?
2. Who is this for?
3. Installation Guide
4. User Guide
5. Credits & License

1. What is this?

The Bcalc Firearm Management App is an open-source firearm, optics, and ammunition tracker with no cloud-based services or subscriptions.

You can enter multiple configurations of firearms, optics, and ammunition. For each configuration, you can track range session performance and conditions. Data reports such as rounds per firearm and top configurations by caliber are available.

An internet connection is not required once installed. No user data is collected or transmitted.

2. Who is this for?

Hunters and hobbyists who prefer to use open-source applications. It's easy to track your firearm performance with different ammunition, whether purchased or hand-load.

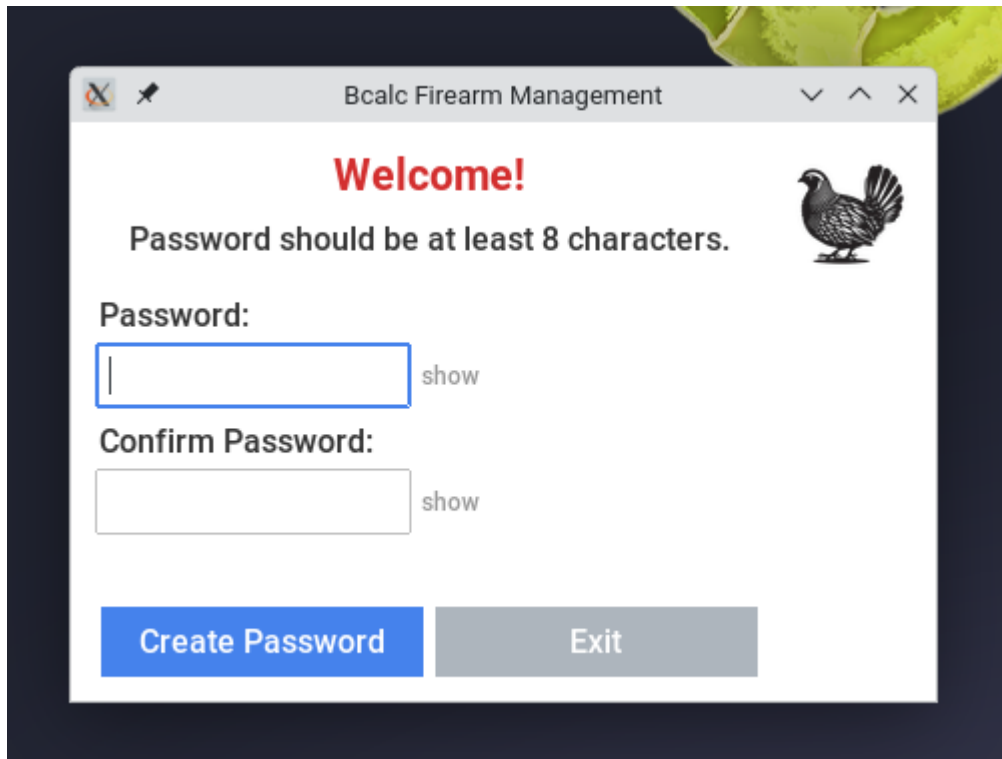
3. Basic Linux Installation Guide

1. Linux (standalone application – no python installation required)
 - download "bcalctracker" from the <https://github.com/briancalc/bcalctracker/releases> page
 - open a terminal in the download folder and make the file executable using `chmod +x bcalctracker`
 - to run using file manager, double-click bcalctracker
 - to run from the terminal `./bcalctracker`
2. Windows
 - in progress
3. Mac
 - in progress

4. User Guide

Step 1. Create a password to secure your records

Don't misplace your password as there is no recovery option.

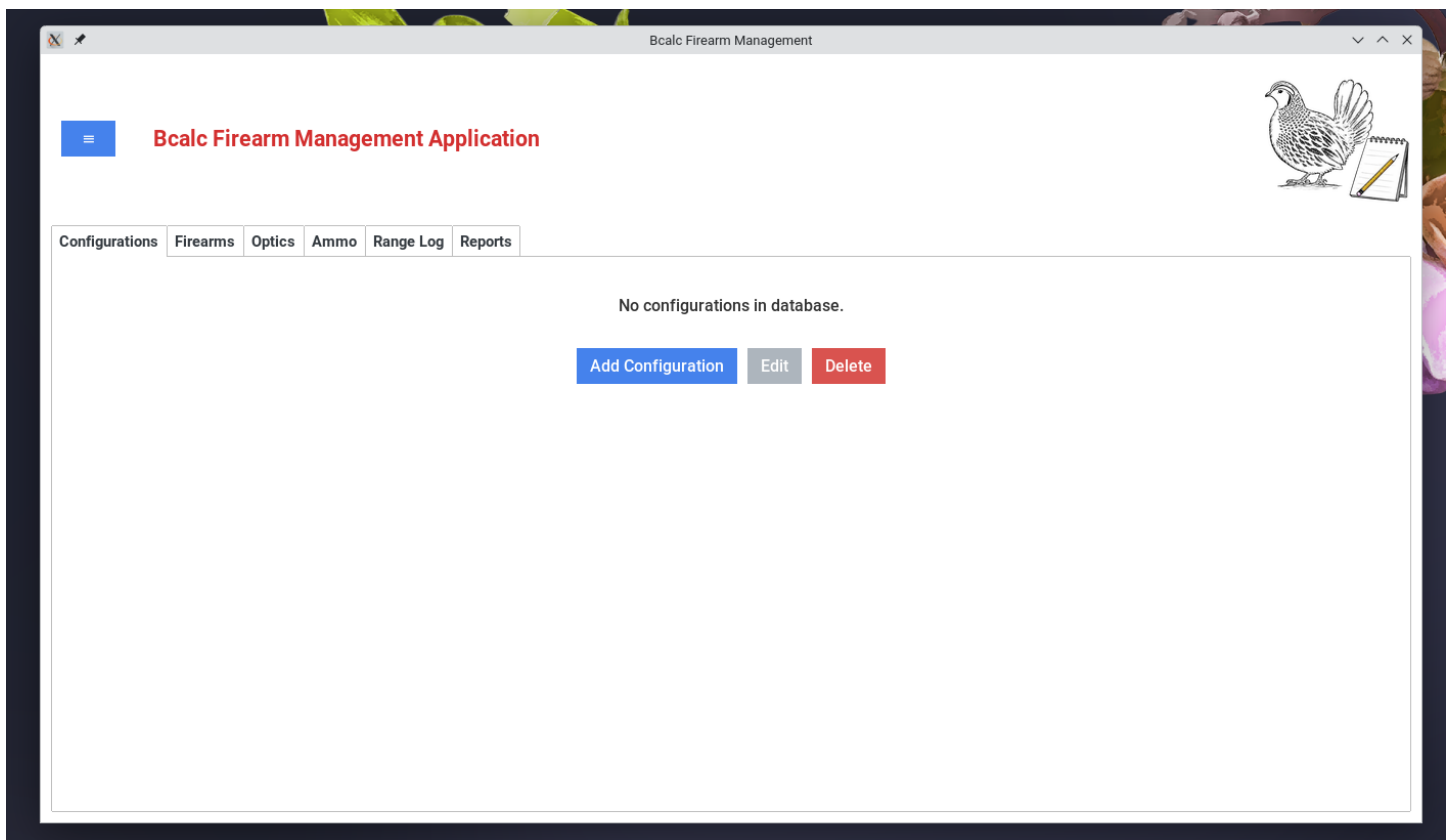


The screenshot shows a web browser window titled "Bcalc Firearm Management". The page has a dark blue background with a green leaf pattern in the top right corner. The main content area is white and contains the following elements:

- A red "Welcome!" heading.
- A black chicken icon.
- A message: "Password should be at least 8 characters."
- A "Password:" label followed by a text input field and a "show" link.
- A "Confirm Password:" label followed by a text input field and a "show" link.
- Two buttons at the bottom: a blue "Create Password" button and a grey "Exit" button.

Step 2: Firearms

Configurations are what facilitate the collection of range session data. Configurations can consist of a firearm, an optic, and a specific ammunition or, at the minimum, just a firearm.



The screenshot shows the main dashboard of the "Bcalc Firearm Management Application". The page has a dark blue background with a green leaf pattern in the top right corner. The main content area is white and contains the following elements:

- A blue sidebar menu with a hamburger icon.
- A red "Bcalc Firearm Management Application" heading.
- A black chicken icon.
- A navigation bar with tabs: "Configurations", "Firearms", "Optics", "Ammo", "Range Log", and "Reports".
- A message: "No configurations in database."
- Three buttons: a blue "Add Configuration" button, a grey "Edit" button, and a red "Delete" button.

So, you need to enter a firearm: click on the Firearms tab and select the blue Add Firearm button. Firearm Name, Caliber Primary, and Barrel Length are required fields. Enter as many of the other fields as you would like, including attaching a photo. You can always return and edit the information later.

The screenshot displays the 'Bcalc Firearm Management' application window. A modal form titled 'Add New Firearm' is open, allowing users to input details for a new firearm. The form is organized into several sections: 'Basic Information' for core details like name, type, and caliber; 'Purchase Details (optional)' for acquisition information; 'Sold Status' for tracking; and 'Photo & Notes' for documentation. The 'Firearm Name' field is pre-filled with 'BAR Mk3 Stalker'. The 'Manufacturer' dropdown is set to 'Browning', and the 'Serial #' field also contains 'BAR Mk3 Stalker'. The 'Caliber Primary' dropdown is set to '243 Win'. The 'Type' dropdown is set to 'Rifle: Semi'. The 'Barrel Length (in)' field is empty. The 'Purchase Date' field is pre-filled with 'YYYY-MM-DD', and the 'Purchase Location' field is empty. The 'Purchase Cost' field is empty. The 'Mark as Sold' checkbox is unchecked. The 'Firearm Photo' field is empty, and the 'Notes' field is empty. A 'Browse...' button is next to the photo field. At the bottom of the modal are 'Save' and 'Cancel' buttons. Below the modal, at the bottom of the application window, are 'Add Firearm', 'Edit', and 'Delete' buttons. The background of the application window shows a sidebar with 'Configurations' and a list of items, and a main area with a decorative image of a bird and a flower.

Add New Firearm

Basic Information

Firearm Name: BAR Mk3 Stalker **Manufacturer:** Browning

Type: Rifle: Semi **Serial #:** BAR Mk3 Stalker

Caliber Primary: 243 Win **Caliber Secondary:**

Barrel Length (in):

Purchase Details (optional)

Purchase Date: YYYY-MM-DD **Purchase Location:**

Purchase Cost:

Sold Status

☐ Mark as Sold

Photo & Notes

Firearm Photo: [Browse...](#)

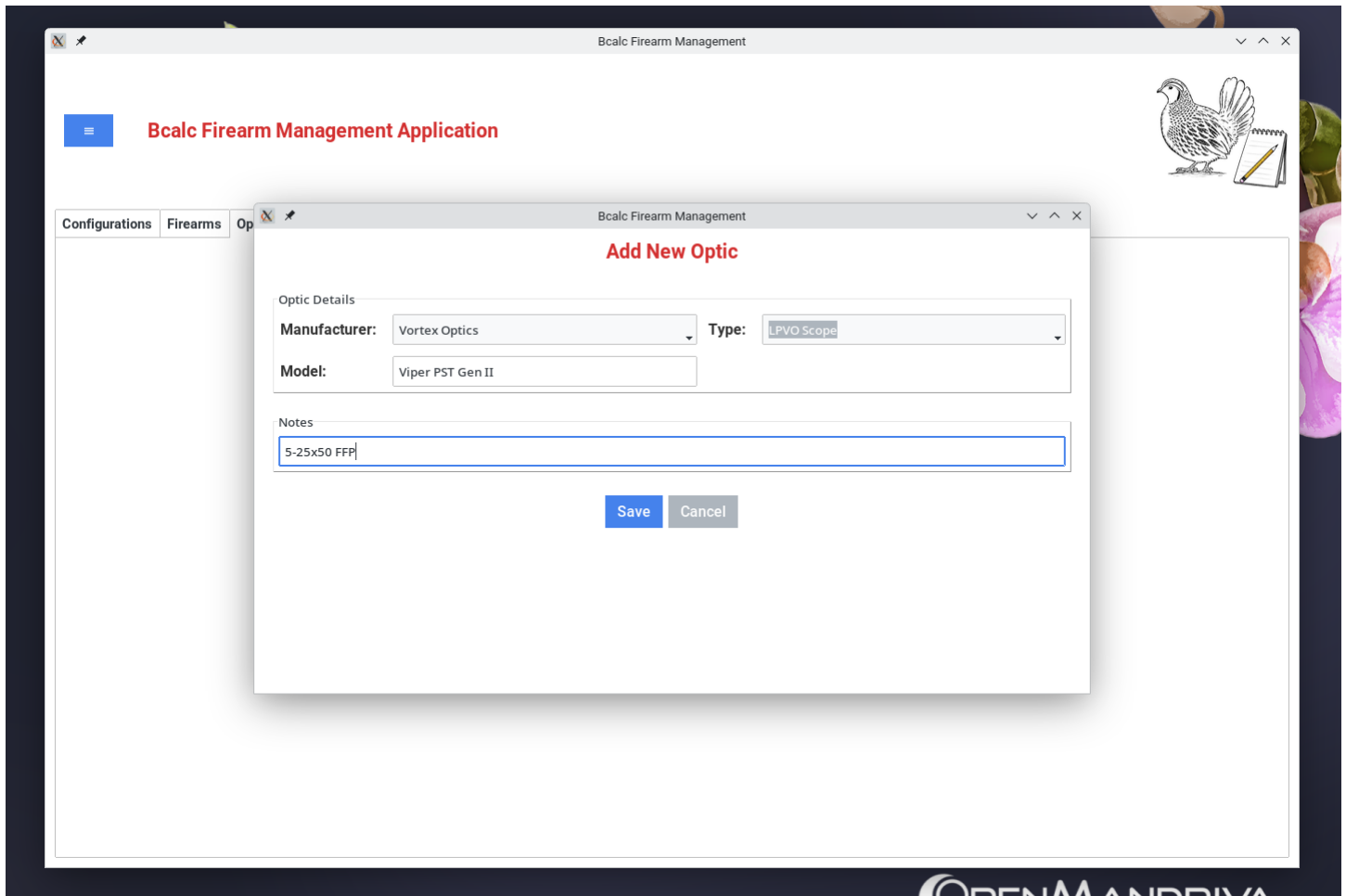
Notes:

[Save](#) [Cancel](#)

[Add Firearm](#) [Edit](#) [Delete](#)

Step 3. Optics

Next, you can enter an optic. Select the Optic Tab and then the Add Optics button.



The screenshot displays the Bcalc Firearm Management Application interface. The main window has a title bar "Bcalc Firearm Management" and a header area with a menu icon and the text "Bcalc Firearm Management Application". Below the header is a navigation bar with tabs labeled "Configurations", "Firearms", and "Optics". The "Optics" tab is currently selected. In the center of the screen, a modal dialog box titled "Add New Optic" is open. This dialog box contains the following fields:

- Optic Details**
 - Manufacturer:** A dropdown menu with "Vortex Optics" selected.
 - Type:** A dropdown menu with "LPVO Scope" selected.
 - Model:** A text input field containing "Viper PST Gen II".
- Notes**
 - A text input field containing "5-25x50 FFP".

At the bottom of the dialog box, there are two buttons: "Save" (in blue) and "Cancel" (in grey). The background of the application window shows a faint illustration of a turkey and some flowers on the right side.

Step 4. Ammunition

You guessed it. Enter ammunition by selecting the Ammo Tab and then the Add Ammo button. Both purchased and hand-load ammunition can be tracked.

Bcalc Firearm Management

Add New Ammo Record

Basic Information

Manufacturer: Caliber/ Gauge:

Grains (Proj.): Projectile Type:

Shot Size: # of Pellets:

Ammo Box Name: Use Case:

Drag Function: Ballistic Coeff.:

☐ Do Not Rebuy (Poor Performance)

Reloading Components (Optional)

Bullet Mfg: Powder Mfg:

Bullet Name: Powder Name:

Brass Mfg: Powder Lot:

Case Weight: Powder Grains:

Overall Length: Primer Mfg:

CBTO: Primer Type:

Crimp Type: Primer Lot:

Hull Mfg: Wad Mfg:

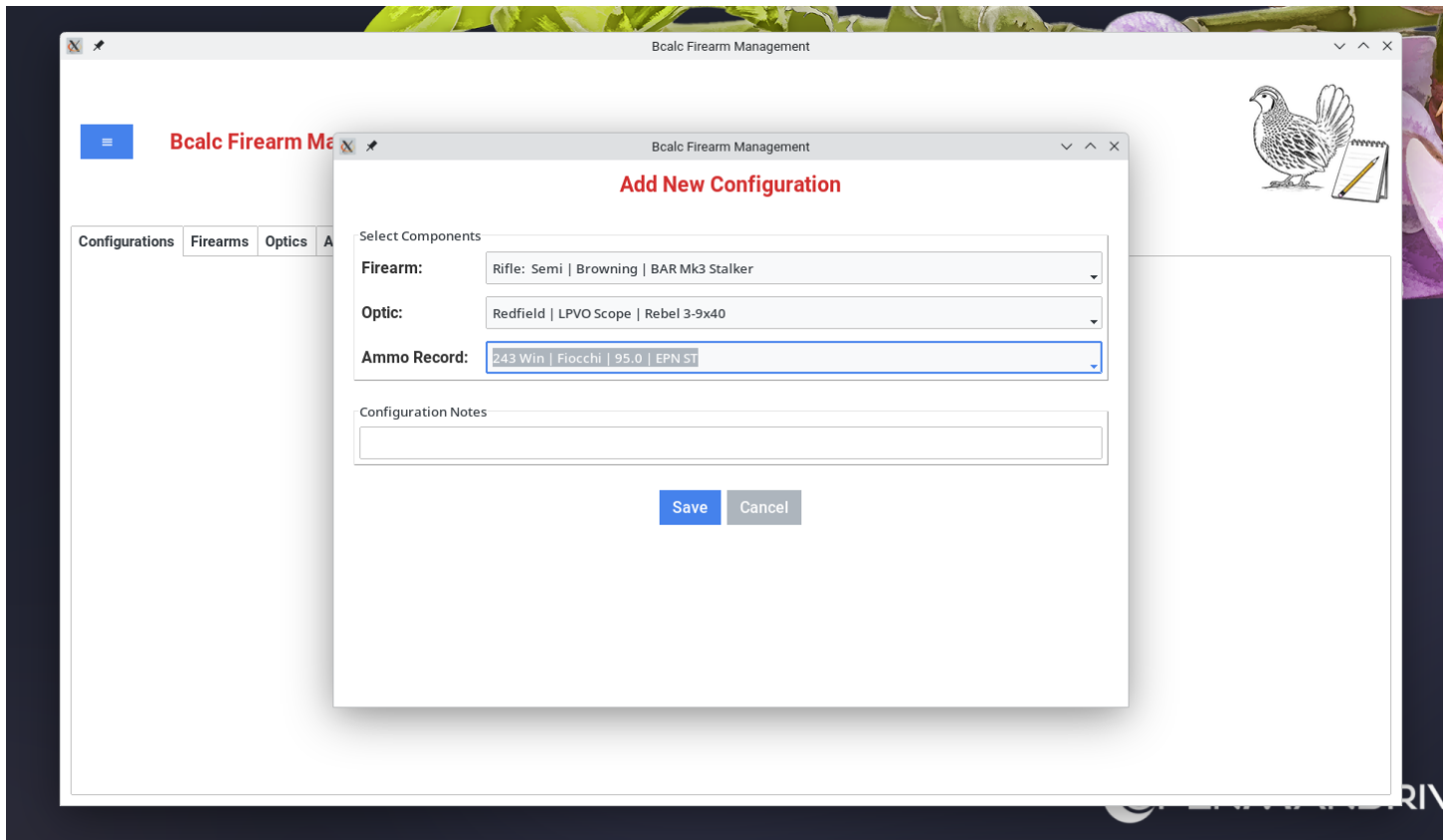
Batch Stamps

Batch Quantity: Date Loaded (YYYY-MM-DD):

Notes

Step 5. Configurations

Enter a configuration by selecting the Configuration Tab and then the Add Configuration button. At a minimum, a configuration must include a firearm. There are no practical limits to the number of different configurations of firearms, optics, and ammunition. It is very likely that each firearm will have several ammo brands associated with it.



The screenshot displays the Bcalc Firearm Management application interface. A modal dialog box titled "Add New Configuration" is open, allowing users to create a new configuration. The dialog includes the following fields:

- Select Components:** A section containing three dropdown menus:
 - Firearm:** Currently set to "Rifle: Semi | Browning | BAR Mk3 Stalker".
 - Optic:** Currently set to "Redfield | LPVO Scope | Rebel 3-9x40".
 - Ammo Record:** Currently set to "243 Win | Fiocchi | 95.0 | EPN ST".
- Configuration Notes:** A text area for additional information.
- Buttons:** "Save" and "Cancel" buttons at the bottom.

The background application window shows a sidebar with tabs for "Configurations", "Firearms", "Optics", and "Ammunition". The "Configurations" tab is active. The application title bar reads "Bcalc Firearm Management". A small illustration of a quail is visible in the top right corner of the application window.

Step 6. Range Log

Select the Range Log Tab and the add Session button. Configurations are the primary key for the range results. All other data is optional. A Rating field is provided to help track the firearm-ammo performance.

Bcalc Firearm Management

Add New Range Session

Session Details

Configuration:

Date (YYYY-MM-DD): Location:

Performance & Condition Data

Rounds Fired:	80	Target Range:	100	Wind Speed:	
Avg Muzzle Vel:	3240	Avg MOA:	1.4	Wind Dir (0-359):	
Max Muzzle Vel:	3260	Best MOA:	1.0	Temperature:	
Std Dev Vel:		Std Dev MOA:		Humidity:	
Ext Spread Vel:		Ext Spread MOA:		Rating:	unknown

Photo & Notes

Target Photo:

Notes:

Step 7. Reports
Select the Reports Tab. Several data points are shown including total rounds per firearm and the top configurations per caliber calculated using MOA and Std Dev MOA data.

Bcalc Firearm Management Application

ConfigurationsFirearmsOpticsAmmoRange LogReports

Firearm Round Stats

Select Firearm: Browning BAR Mk3 Stalker

Total Rounds Fired: 195

Top Ammo: Fiocchi EPN ST (243 Win)

Configuration Stats

Select Configuration: Henry H9 Provider | Leupold | Hornady

Avg Velocity: 2465 fps

Avg MOA: 1.50

Top Configurations by Caliber

Select Caliber: 243 Win

Rank	Configuration	Rating	Rounds
1	Browning BAR Mk3 Stalker Redfield Reb	No Rating	195
2	Ruger American Gen II Predator Vortex	No Rating	85

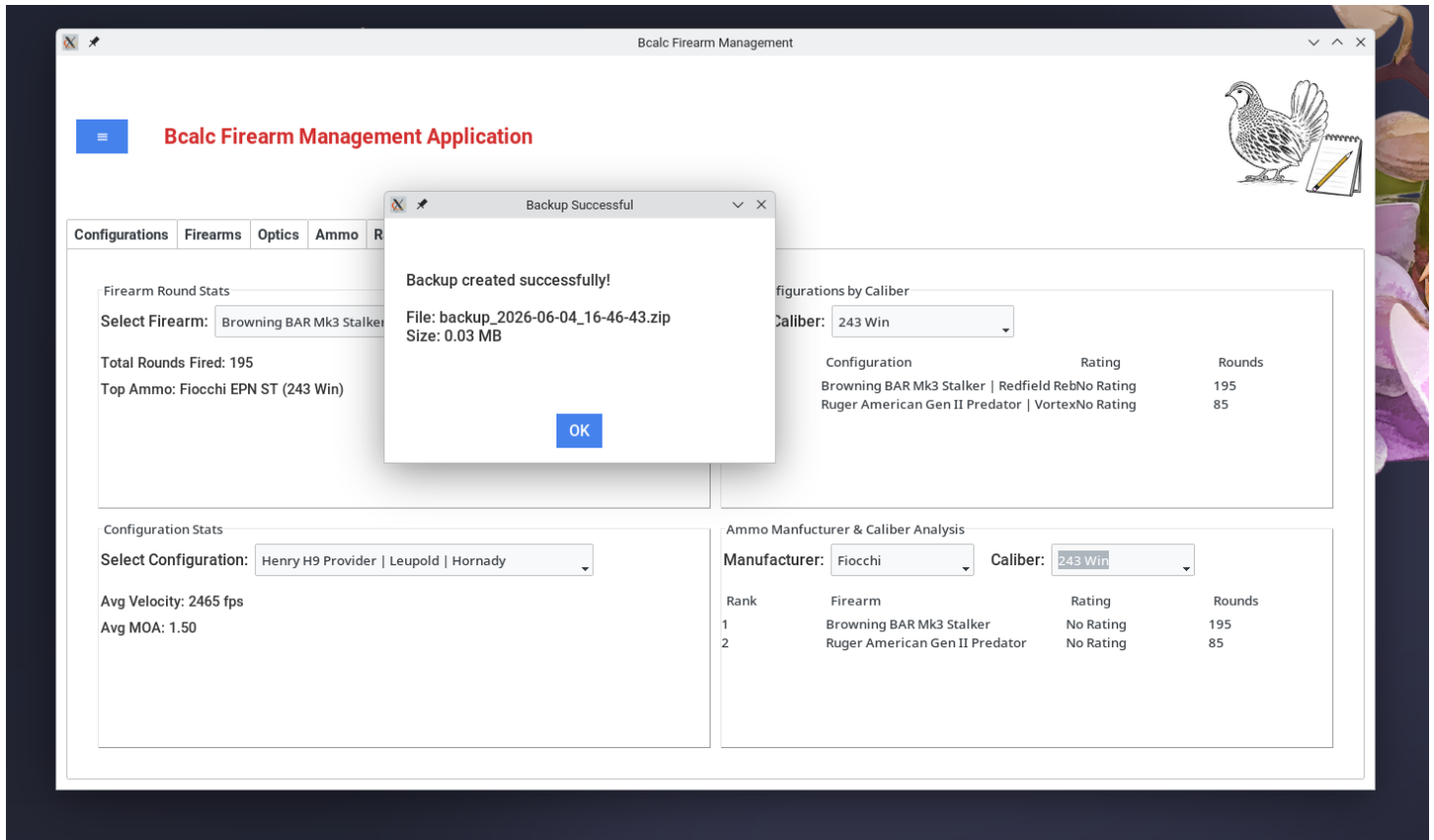
Ammo Manufacturer & Caliber Analysis

Manufacturer: FiocchiCaliber: 243 Win

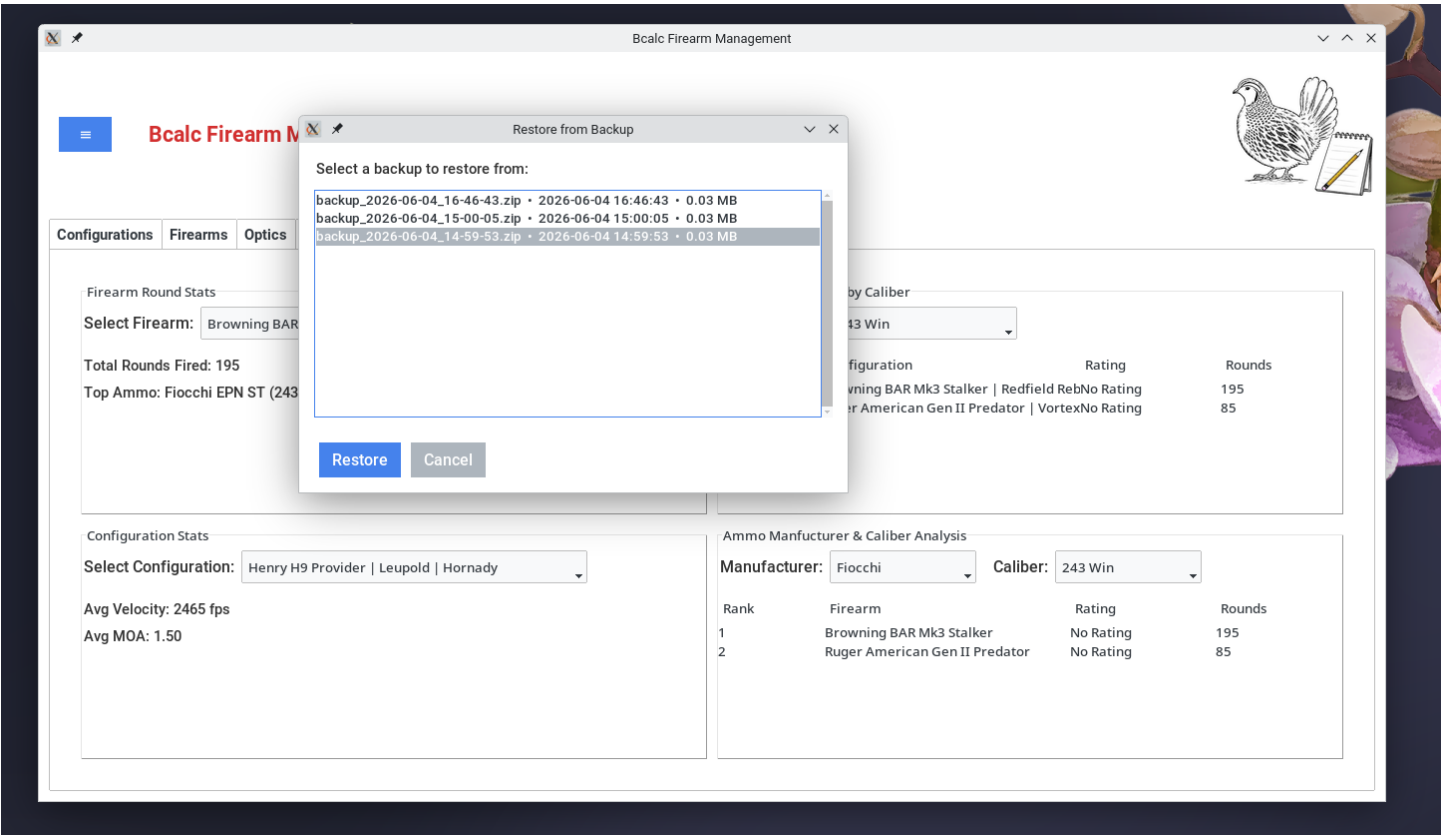
Rank	Firearm	Rating	Rounds
1	Browning BAR Mk3 Stalker	No Rating	195
2	Ruger American Gen II Predator	No Rating	85

Step 8. Optional: Backup and Restore

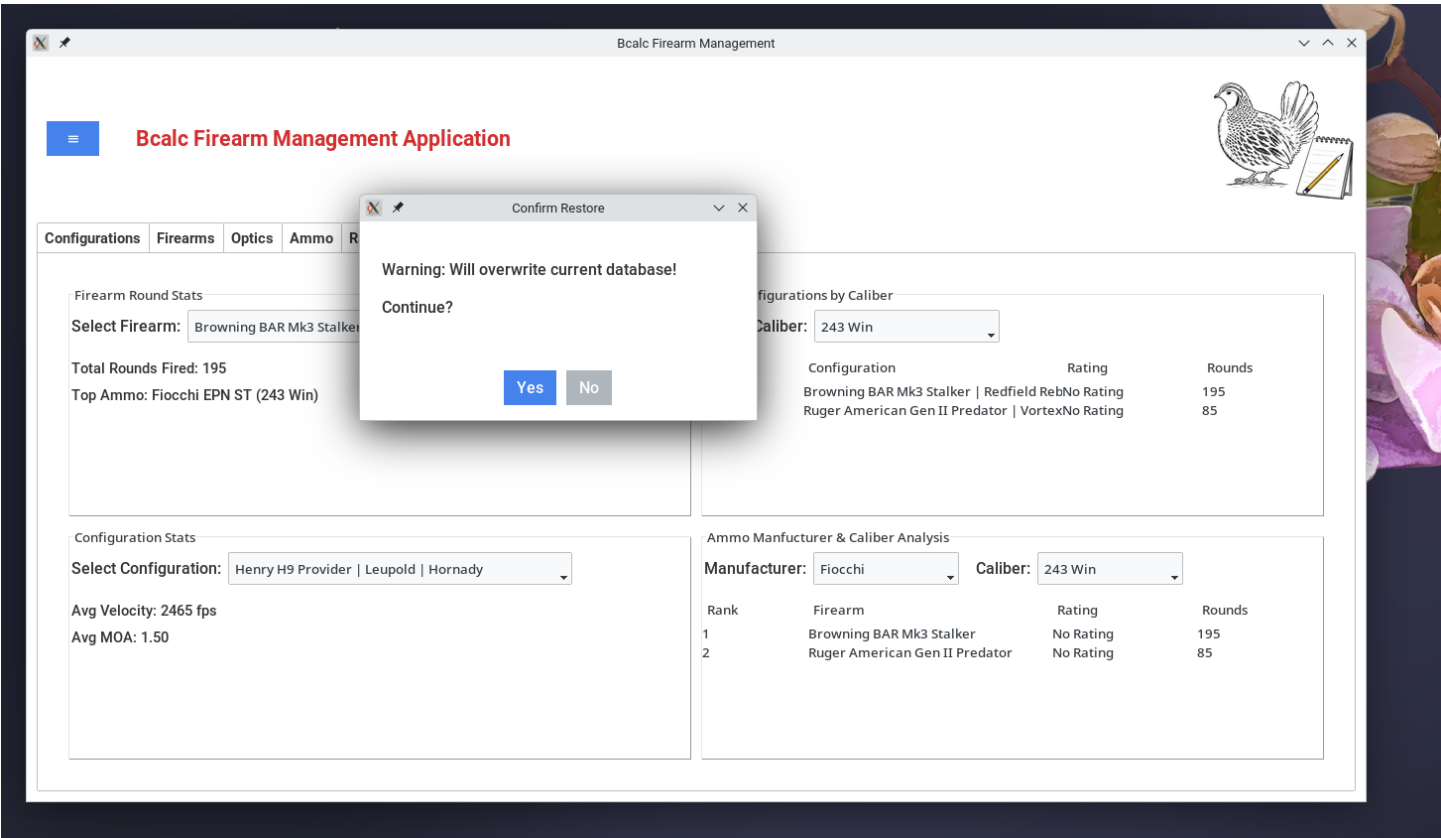
It's a good idea to periodically save a copy of the database just in case. Click the blue hamburger menu at the top left of the window and select Backup Database.



To restore your records from a backup, click the blue hamburger menu and select Restore Database. Pick a backup and press Restore.



Your current database will be replaced by the backup so be certain you want to restore your previous records.



6. Credits & License

I'm not a software developer. The goal is to continue my python learning progress while producing something useful, at least for me. This application is offered for educational and entertainment purposes.

The Bcalc Firearm Management App was created by Brian Calc.

GNU General Public License (GPL) Version 3.